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Adaptive Wavelet-Domain Image Watermarking Using Genetic Algorithm Optimization for Digital Content Protection

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Abstract: Digital watermarking plays an important role in protecting digital content, verifying ownership, and supporting the trustworthy distribution of digital information. However, maintaining watermark robustness without compromising visual quality remains a challenge, particularly when embedding strength is selected through empirical adjustment. This study investigates an adaptive wavelet-domain watermarking method that uses genetic algorithm (GA) optimization to determine embedding energy for digital image protection. The watermark was encrypted using the Data Encryption Standard (DES) and embedded into selected wavelet subbands. A GA was employed to adjust the embedding strength according to an image-distortion criterion, while watermark quality and extraction performance were evaluated using relative error (RE) and other image-quality measures. The proposed method was tested against cropping, rotation, scaling, Joint Photographic Experts Group (JPEG) compression, filtering, Gaussian noise, and other distortions, and its performance was compared with manually selected embedding energy and a least significant bit (LSB) method. The results showed that the GA-based method achieved watermark extraction rates of 87.2% after 25% cropping, 93.4% after 15° rotation, and 96.3% under JPEG compression at a quality factor of 50. Under these conditions, the corresponding extraction rates for manually selected embedding energy were 83.1%, 89.2%, and 91.8%, respectively. The proposed method also maintained low visual distortion in the reported imperceptibility test. These findings indicate that GA-based parameter selection is a feasible alternative to manual adjustment in wavelet-domain watermarking. The method provides a technical approach to improving the resilience of embedded ownership information against common image-processing operations, with potential applications in digital content protection and copyright verification.

Keywords: Digital content protection; Digital watermarking; Genetic algorithm; Adaptive embedding optimization; Wavelet transform; Copyright protection

1 Introduction

The protection of digital information has become an important concern in the storage, transmission, and distribution of digital content. Images, audio recordings, videos, documents, and other digital materials can be reproduced and modified with relative ease, creating difficulties in establishing ownership and maintaining the integrity of distributed content. Information hiding provides a technical means of embedding secret or auxiliary information within an apparently innocuous carrier without making the embedded information readily apparent. The carrier may be an image, an audio signal, a video sequence, a document, or another form of digital data. Once the information has been embedded, the resulting carrier can be transmitted through an open communication channel

while concealing the presence of the hidden message. Unlike conventional encryption, which transforms readable information into ciphertext, information hiding primarily seeks to conceal the existence of the embedded information. This distinction is particularly relevant in communication environments where visibly encrypted data may attract attention or become subject to additional scrutiny.

Cryptographic techniques protect digital information by converting plaintext into ciphertext that cannot readily be interpreted without the appropriate decryption key. Nevertheless, encrypted information remains identifiable as protected data, even when its contents cannot be recovered. Historical accounts of concealed communication indicate that hiding the existence of information has long been used when openly transmitting a protected message might raise suspicion [1]. Similarly, communication between individuals or organizations already subject to observation may attract further attention when the transmitted material consists entirely of recognizable ciphertext [2]. Information hiding addresses a different aspect of information protection by concealing the presence of embedded content rather than relying exclusively on encryption. The two approaches are therefore complementary: encryption restricts access to information, whereas information hiding makes the presence of that information less apparent. Their combination is relevant to digital communication systems in which both confidentiality and unobtrusive information transmission are required.

Digital watermarking is a related information-hiding technique with particular relevance to digital content protection, copyright verification, ownership identification, authentication, integrity assessment, and content tracking. By embedding identifying information directly into digital media, watermarking provides a means of associating a digital object with information about its origin or ownership. This association may remain available after the content has been copied, transmitted, or subjected to certain modifications, making watermarking useful in digital content management and intellectual property protection. Although digital watermarking and information hiding share fundamental embedding principles, their objectives are not identical. Information hiding generally emphasizes concealing the existence of secret information, whereas digital watermarking places greater emphasis on the continued availability and reliable recovery of embedded information under specified operating conditions. Watermarking systems must therefore satisfy application-dependent requirements concerning visual quality, robustness, security, and extraction reliability. In this study, the watermark represents the information embedded within a cover image, while the principles of information hiding provide the technical basis for its insertion and subsequent extraction.

Wavelet-domain watermarking has received considerable attention because the discrete wavelet transform (DWT) provides a multiresolution representation of image information [3]. By decomposing an image into frequency subbands with different spatial characteristics, the DWT allows watermark information to be inserted selectively into regions that differ in perceptual sensitivity and resistance to image processing. This property is particularly useful when a watermark must remain recoverable without introducing noticeable changes to the protected image. Wavelet-domain watermarking can also operate alongside cryptographic protection: encryption restricts unauthorized interpretation of the watermark content, while embedding maintains an association between the watermark and the host image. However, the practical value of this association depends on whether the watermark can still be recovered after the image undergoes compression, filtering, geometric transformation, or deliberate manipulation. For digital ownership verification and content protection, the reliability of watermark extraction is therefore as important as the initial embedding operation. Several requirements must be considered when designing a watermarking system for such applications.

(1) Imperceptibility. Imperceptibility describes the extent to which embedding a watermark leaves the visual appearance of the host image unchanged under ordinary viewing conditions. The embedding operation inevitably modifies some image information, but the resulting distortion should remain sufficiently small to preserve the perceptual quality of the protected content. This requirement is particularly important when watermarked images are intended for publication, distribution, or subsequent processing. The term imperceptibility is more precise than invisibility in this context because the watermark remains physically represented in the image data even when its presence cannot readily be distinguished by a human observer. Imperceptibility can be assessed using quantitative image-quality measures, although numerical similarity alone does not fully describe human visual perception.

(2) Robustness. Robustness refers to the ability of an embedded watermark to remain detectable or recoverable after the host image has undergone intentional or unintentional modifications. A robust watermark should survive a specified range of image-processing operations and resist removal without substantial degradation of the protected content. When an attacker has only partial knowledge of the watermarking process, including limited information about the location or structure of the embedded watermark, successful removal should ideally require modifications that noticeably affect the host image [4]. Robustness is particularly important for copyright verification because digital content may pass through multiple processing and distribution stages before ownership information is examined. The resistance of a watermarking method is commonly assessed against several categories of attacks.

• Common signal-processing operations. These operations include digital-to-analogue and analogue-to-digital conversion, resampling, requantization, dithering, recompression, contrast adjustment, colour modification, and other routine image-processing procedures [5]. Although such operations may be performed for legitimate purposes,

they can modify the image coefficients or pixel values that carry the embedded watermark. A watermark intended for practical digital content protection should remain recoverable after a reasonable range of these transformations.

• Geometric distortions. Geometric attacks modify the spatial arrangement of image information through operations such as rotation, translation, cropping, and scaling [6, 7]. Unlike some signal-processing attacks, geometric transformations can disrupt the correspondence between the original embedding positions and the locations examined during watermark extraction. Even relatively small changes in image geometry may therefore reduce extraction reliability. Resistance to these distortions is important when protected images are resized, reformatted, cropped, or otherwise modified during digital distribution.

• Collusion and forgery attacks. Collusion attacks may occur when individuals possessing different watermarked copies combine or manipulate those copies in an attempt to weaken or remove the embedded information. Forgery attacks involve the creation or alteration of watermark information to produce a misleading indication of ownership or authenticity. Such attacks are especially relevant when digital watermarks are used as supporting evidence in copyright disputes or ownership verification [8, 9]. A watermarking system intended for these applications should account for the possibility of deliberate manipulation, although protection against collusion and forgery requires security considerations beyond resistance to ordinary image distortion.

(3) Security. Security concerns the resistance of the watermarking system to intentional attempts to identify, analyse, modify, remove, or forge embedded information. Although security is closely related to robustness, the two properties address different aspects of system performance. Robustness describes the survival of watermark information under specified transformations, whereas security additionally considers whether an adversary can exploit knowledge of the algorithm, embedding locations, keys, or statistical properties of the watermarked image. Ideally, watermark insertion should not introduce easily identifiable statistical patterns, and unauthorized removal should not be possible without significant modification of the protected content [10]. In systems that encrypt watermark information before embedding, the security of the encryption procedure and the robustness of the embedding mechanism must also be considered separately.

(4) Unambiguous detection or extraction. A watermarking system must support reliable identification or reconstruction of the embedded information. This requirement is especially important when the watermark is used to establish ownership, verify authenticity, or associate digital content with a particular source [11, 12]. An extraction procedure should minimize ambiguity between the original watermark and the recovered information, including after the host image has undergone permissible modifications. The quality of the extracted watermark therefore provides an important measure of the practical reliability of the embedding method.

(5) Graceful degradation. Watermark extraction performance should ideally decline progressively as the severity of image distortion increases rather than fail completely after a relatively small disturbance [13]. A watermark may remain recognizable even when some embedded information has been altered or lost. Such behaviour is particularly relevant to digital content that may experience repeated compression, geometric modification, or other processing operations during distribution. The ability to retain partial recognition under increasingly adverse conditions provides additional information about robustness beyond a simple distinction between successful and unsuccessful extraction.

Among these requirements, imperceptibility and robustness are closely associated with the energy allocated to watermark embedding. Increasing the embedding strength can improve resistance to certain image-processing operations by making the watermark less vulnerable to coefficient modification or information loss. However, excessive embedding energy may alter image structure and produce visible distortion, reducing the quality of the protected content. Conversely, reducing embedding energy generally preserves image fidelity but may increase the likelihood of watermark degradation during compression, filtering, or geometric transformation. The appropriate embedding strength therefore depends on the relationship between the perceptual characteristics of the host image, the embedding domain, and the distortions that the watermarked image is expected to withstand. In wavelet-domain systems, this relationship is further influenced by the frequency subbands selected for embedding. Determining an appropriate embedding energy is consequently a constrained parameter-selection problem involving competing requirements rather than a simple attempt to maximize watermark strength.

A central difficulty in wavelet-domain watermarking is the selection of embedding energy that provides adequate resistance to image manipulation while keeping visual distortion within an acceptable range. In conventional implementations, the embedding parameter is frequently selected through empirical adjustment. Several candidate values are tested, and the final parameter is chosen according to the observed compromise between image quality and watermark extraction performance. Although this approach can produce acceptable results, it depends on the choice of candidate values and the experience of the researcher. It may also require repeated embedding and evaluation operations, particularly when different image characteristics or attack conditions are considered. More importantly, a parameter selected through manual experimentation cannot necessarily be regarded as optimal under a clearly defined evaluation criterion. The relationship between embedding strength, perceptual distortion, and resistance to image manipulation may be nonlinear, making systematic parameter selection preferable to repeated trial-and-error adjustment. For digital content protection applications, this issue is relevant because inconsistent embedding settings

may lead to different levels of watermark reliability across protected images.

Adaptive optimization provides a means of reducing dependence on manually selected embedding parameters. Genetic algorithms (GAs) are population-based search methods that examine candidate solutions through iterative selection, crossover, and mutation. Their ability to explore parameter spaces without requiring an explicit derivative of the evaluation function makes them applicable to nonlinear optimization problems. In digital watermarking, a GA can be used to search for an embedding parameter that satisfies a specified image-quality criterion while operating within the constraints imposed by the watermarking scheme. The resulting parameter can then be evaluated in terms of watermark recovery after different image-processing operations. However, the quality of the selected solution depends on the fitness function, parameter bounds, and stopping criteria used during optimization. Automated parameter selection alone does not guarantee improved robustness; its practical value must be established through comparison with conventional embedding methods under corresponding experimental conditions.

This study investigates an adaptive wavelet-domain image watermarking method in which watermark encryption, frequency-domain embedding, and GA-based embedding-parameter selection are combined within a single procedure. The watermark was first encrypted using the Data Encryption Standard (DES) before being inserted into the host image [14]. Encryption was included to restrict direct interpretation of the embedded watermark data, while the embedding procedure was responsible for maintaining its recoverability under image modification. Following DWT decomposition, the encrypted watermark information was embedded primarily into selected middle-frequency subbands, with the embedding strategy also considering lower-frequency components to improve resistance to distortion. These subbands were selected to balance the relative stability of the embedded information against the visual sensitivity of the host image. A GA was then applied to determine the embedding strength automatically according to the specified distortion-based evaluation criterion. The method thus replaced repeated manual adjustment with an iterative parameter-search procedure while retaining the wavelet-domain structure of the watermarking system.

The proposed watermarking method was examined through imperceptibility and robustness experiments involving several forms of image processing and geometric distortion. Watermark extraction was evaluated after cropping, shearing, rotation, scaling, Joint Photographic Experts Group (JPEG) compression, filtering, Gaussian noise, and mosaic-based attacks. Relative error (RE) was used to assess image differences and watermark reconstruction quality, while additional image-quality measures were considered in the evaluation of perceptual distortion. The experiments also compared GA-selected embedding energy with manually selected embedding energy and a conventional least significant bit (LSB) watermarking method. These comparisons were intended to examine whether automated parameter selection provided a more consistent compromise between image fidelity and watermark recovery under the tested conditions. The resulting observations were considered in relation to the technical requirements of digital image protection and the continued availability of embedded ownership information after image modification.

The main contribution of this study lies in the implementation and evaluation of a GA-based embedding-energy selection procedure within an encrypted wavelet-domain image watermarking system. The proposed method combines DES-based watermark preparation, frequency-selective embedding, and adaptive parameter selection to reduce dependence on empirical adjustment of embedding strength. Its performance is examined using both visual-quality criteria and watermark extraction measurements under a range of image-processing and geometric attacks. From an application perspective, the method addresses a practical requirement of digital content protection: embedding identifying information in digital images without introducing excessive visible distortion while retaining recoverability after common modifications. The study therefore provides an experimental basis for assessing adaptive wavelet-domain watermarking as a technical approach to supporting copyright verification and ownership identification in digital content distribution. The following sections describe the wavelet transform, watermark embedding and extraction procedures, GA-based parameter selection, and experimental results.

2 Wavelet Transform

The wavelet transform is a well-established mathematical tool for time-frequency analysis and multiresolution signal representation [15]. Its ability to characterize signal components at different spatial and frequency scales makes it particularly suitable for digital image processing, where structural information, edges, textures, and other image features are distributed across different frequency bands. The theoretical development of wavelet-based multiresolution analysis during the 1980s established an important foundation for subsequent applications in signal decomposition and reconstruction [16]. In digital watermarking, multiresolution decomposition allows embedded information to be allocated to selected frequency components according to their relative perceptual sensitivity and resistance to image manipulation. This property is relevant to the protection of digital images because the embedding procedure must preserve visual quality while retaining sufficient information for reliable watermark extraction.

Let $f(x)$ be a continuous square-integrable function, and let $\psi(x)$ denote an admissible mother wavelet. The

continuous wavelet transform (CWT) of $f(x)$, denoted by $WT_f(a,\tau)$, is defined as follows [14]:

$$WT_f(a,\tau) = \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} f(t)\psi^* \left(\frac{t-\tau}{a} \right) dt \tag{1}$$

where, $a > 0$ is the scale parameter, τ is the translation parameter, and the superscript $*$ denotes complex conjugation. The scale parameter determines the frequency range represented by the wavelet, whereas the translation parameter specifies its position in the signal. Smaller scale values generally correspond to finer signal details, while larger values represent broader variations.

The scaled and translated wavelet function is expressed as

$$\psi_{a,\tau}(x) = \frac{1}{\sqrt{a}} \psi \left(\frac{x-\tau}{a} \right) \tag{2}$$

Provided that the admissibility and reconstruction conditions are satisfied, the original function can be recovered from its continuous wavelet coefficients through the inverse transform:

$$f(x) = \frac{1}{C_\psi} \int_0^\infty \int_{-\infty}^\infty WT_f(a,\tau) \frac{\psi_{a,\tau}(x)}{a^2} d\tau \, da \tag{3}$$

where, C_ψ is the wavelet admissibility constant, defined by

$$C_\psi = \int_{-\infty}^{+\infty} \frac{|\hat{\psi}(u)|^2}{|u|} du < \infty \tag{4}$$

Here, $\hat{\psi}(u)$ represents the Fourier transform of $\psi(x)$. The admissibility condition requires C_ψ to be finite and, under the usual regularity assumptions, implies $\hat{\psi}(0) = 0$. This condition is associated with the zero-mean property of the wavelet and permits the analysis of signal variations across different scales. For a real-valued mother wavelet, the complex conjugation in the transform and reconstruction expressions has no effect on the resulting values.

For digital image processing, the DWT provides a computational representation of multiresolution decomposition. In a dyadic wavelet framework, the analysis scales follow powers of two, allowing the signal to be represented through frequency bands with progressively changing resolution. Higher-frequency components describe relatively fine variations in the signal, whereas lower-frequency components represent broader structural information. Such a decomposition provides a means of separating image details from more slowly varying image characteristics. The filter-bank principle associated with Mallat's multiresolution framework uses low-pass and high-pass filtering to obtain approximation and detail information at successive scales.

In the dyadic recursive representation adopted in the original formulation, the approximation and detail components of a discrete signal $f(n)$ are expressed as

$$S_{2^j} f(n) = \sum_{k \in \mathbb{Z}} h_k S_{2^{j-1}} f(n - 2^{j-1}k) \tag{5}$$

and

$$W_{2^j} f(n) = \sum_{k \in \mathbb{Z}} g_k S_{2^{j-1}} f(n - 2^{j-1}k) \tag{6}$$

where, $S_{2^j} f(n)$ and $W_{2^j} f(n)$ denote the approximation and detail components at scale 2^j, respectively. The coefficients h_k and g_k correspond to the low-pass and high-pass analysis filters. The low-pass filter retains the slowly varying components of the signal, while the high-pass filter extracts variations associated with finer details. Successive application of these filters generates a multiscale representation of the original signal.

The original study specifies a second-order spline wavelet basis for image decomposition. The corresponding filter coefficients used in the reported formulation are retained as follows:

$$h_0 = 0.375, \quad h_1 = 0.375, \quad h_2 = 0.125, \quad h_3 = 0.125 \tag{7}$$

$$g_0 = -2.0, \quad g_1 = 2.0, \quad g_2 = 0, \quad g_3 = 0 \tag{8}$$

These coefficients are specified for $k = 0, 1, 2, 3$, with

$$h_k = g_k = 0, \quad k > 3 \tag{9}$$

The low-pass coefficients determine the contribution of neighbouring signal samples to the approximation component, whereas the high-pass coefficients govern the extraction of local variations. The resulting decomposition

separates signal information across different scales and provides the frequency-domain representation used for subsequent watermark embedding.

Through successive dyadic decomposition, an image can be represented by components associated with different spatial-frequency ranges. In the scale interpretation adopted here, relatively fine image details are associated primarily with the smaller scales, including $S = 2^1$ and $S = 2^2$. Such components commonly contain edges, fine textures, and high-frequency noise. Their sensitivity to filtering, compression, and other image-processing operations can limit their suitability for robust watermark embedding. Components at larger scales, including $S = 2^3$, $S = 2^4$, and above, describe progressively coarser image structures and lower-frequency variations [17]. These components generally retain more of the large-scale image information, although modifying them may introduce perceptually noticeable distortion.

For two-dimensional images, wavelet decomposition is applied along the horizontal and vertical directions to obtain subbands with different frequency characteristics. These subbands provide the basis for selecting embedding locations according to the required balance between visual quality and watermark recovery. In the proposed watermarking framework, a two-dimensional spline wavelet basis was used throughout the image decomposition process. The resulting frequency-domain representation supported the embedding of encrypted watermark information into selected subbands, with the embedding strength subsequently adjusted using the genetic algorithm described in Section 3. This arrangement allows the wavelet transform to serve as the signal-processing foundation of the proposed digital content protection method.

3 Watermark Insertion and Extraction Strategy

Information hiding provides a means of embedding information within digital media without making its presence readily apparent to an unauthorized observer. In digital watermarking, this principle is applied to associate identifying information with protected content while preserving its visual characteristics. The embedded watermark should remain difficult to perceive during normal viewing and sufficiently recoverable after common image-processing operations. These requirements are particularly important in digital content protection, where images may undergo compression, redistribution, or modification before their ownership information is examined.

The effectiveness of a watermarking system depends on both the structure of the embedded information and the strategy used to insert it into the host image. Embedding strength is particularly important because it directly affects image distortion and watermark recoverability. Excessively weak embedding may result in substantial information loss after image processing, whereas excessive embedding energy can produce visible changes in the protected image. To address this problem, the proposed method combined wavelet-domain watermark embedding with GA-based parameter selection. The GA was used to determine the embedding strength according to a specified distortion criterion, reducing the need for repeated manual adjustment. The resulting watermarking procedure was evaluated in terms of visual quality and resistance to image-processing and geometric attacks.

The selection of appropriate frequency subbands is another important consideration in watermark embedding. High-frequency components contain fine image details and are particularly sensitive to operations such as compression, filtering, and noise reduction. Information embedded exclusively in these components may therefore be lost when the image undergoes routine processing. Low-frequency components, by contrast, retain much of the image's overall structural information and may provide greater resistance to certain forms of distortion. However, substantial modification of low-frequency coefficients can affect visually prominent image features because the human visual system is sensitive to changes in broad image structures. Embedding strategies must therefore account for the different characteristics of the frequency subbands rather than treating all wavelet coefficients as equally suitable for watermark insertion.

Considering these characteristics, the proposed method adopted a frequency-selective embedding strategy in which the encrypted watermark was primarily inserted into the middle-frequency wavelet subbands, particularly LH and HL. Here, L and H represent low- and high-frequency components, respectively, while LL, LH, HL, and HH denote the low-low, low-high, high-low, and high-high subbands obtained through two-dimensional wavelet decomposition. The middle-frequency subbands were selected because they provide a practical compromise between preserving image quality and maintaining watermark information after common processing operations. The embedding strength was adjusted through GA-based parameter selection and evaluated using the image-quality criteria described in Section 3.1. This frequency-selective procedure was intended to preserve the visual characteristics of the cover image while retaining watermark information under the investigated image-processing conditions.

3.1 Image Quality Evaluation Criteria

The strength of watermark embedding directly affects both image quality and the recoverability of embedded information. Lower embedding energy generally produces less distortion in the cover image but may reduce watermark resistance to subsequent image processing. Increasing embedding energy can improve watermark recovery

under certain attack conditions, although excessive modification of image coefficients may introduce noticeable visual distortion. Quantitative measures are therefore required to evaluate the relationship between embedding strength and image quality.

Three measures were adopted in this study: signal-to-noise ratio (SNR), peak signal-to-noise ratio (PSNR), and RE. Let the original cover image and the corresponding watermarked or distorted image be represented by two grayscale matrices of identical dimensions, $M \times N$. These matrices contain the pixel intensities of the original and processed images, respectively. The following criteria were used to quantify the differences between them.

3.1.1 Criteria for the cover image

- SNR:

$$\mathrm{SNR} = 10 \log_{10} \frac{\sum_{i=1}^{M} \sum_{j=1}^{N} a_{ij}^2}{\sum_{i=1}^{M} \sum_{j=1}^{N} (a_{ij} - b_{ij})^2} \tag{10}$$

- PSNR:

$$\mathrm{PSNR} = 10 \log_{10} \frac{b_{\max}^2}{\mathrm{MSE}} \tag{11}$$

where, $b_{\max}$ is the maximum of the gray value of image pixel, and MSE is the mean-square error:

$$\mathrm{MSE} = \frac{1}{M \times N} \sum_{i=1}^{M} \sum_{j=1}^{N} (a_{ij} - b_{ij})^2 \tag{12}$$

- RE:

$$\mathrm{RE} = \frac{\sqrt{\sum_{i=1}^{M} \sum_{j=1}^{N} (a_{ij} - b_{ij})^2}}{\sqrt{\sum_{i=1}^{M} \sum_{j=1}^{N} (a_{ij})^2}} \tag{13}$$

The larger the RE, the larger the difference between the watermarked image and the cover image. In general, when $\mathrm{RE} < 5\%$, the difference is visually imperceptible. The 5% criterion applies to the no-attack scenario. Under attack, a higher RE (e.g., up to 15%) may still be tolerable if the extracted watermark remains recognizable. In this paper, PSNR, SNR, and RE are used together as criteria for evaluating image quality.

3.1.2 Criterion for the extracted watermark

The success rate of watermark extraction (SRWE) is defined as:

$$\mathrm{SRWE} = 1 - \mathrm{RE} \tag{14}$$

where, RE denotes the relative error between the original watermark and the extracted watermark. SRWE is expressed as a percentage and represents the reconstruction quality of the extracted watermark according to the specified relative-error measure. When $\mathrm{RE} = 0$, $\mathrm{SRWE} = 100\%$, indicating that no reconstruction error is detected. When $\mathrm{RE} = 1$, $\mathrm{SRWE} = 0\%$. A higher SRWE indicates closer agreement between the original and extracted watermark images. This measure should not be interpreted as the probability of successful ownership identification.

3.1.3 Channel splitting and criteria for color images

An RGB colour image consists of three channels representing red (R), green (G), and blue (B) intensity information. For an 8-bit image, each channel contains pixel values ranging from 0 to 255. In the proposed procedure, the RGB channels were separated and treated as individual grayscale matrices before subsequent wavelet-domain processing.

For an original colour image of size $M \times N$, each of the three channel matrices has the same spatial dimensions. The matrices were concatenated horizontally in the order R, G, and B to obtain a single grayscale representation of size $M \times 3N$. This representation retained the pixel information from all three colour channels while allowing the watermarking procedure to operate on a two-dimensional matrix.

The resulting matrix was processed using the same wavelet decomposition and image-quality evaluation criteria applied to grayscale images. After the corresponding operations, the channel information could be separated and rearranged into the original RGB configuration. This representation was used to handle the colour-image experiments within the proposed watermarking framework.

3.2 Watermark Structure

The structure of the embedded watermark affects its recoverability after image processing. Fine details in a watermark image may be particularly sensitive to noise, smoothing, and other operations that alter high-frequency information. The watermark data were therefore converted into a one-dimensional sequence and encrypted using the DES before embedding. Encryption was applied to the watermark data rather than to the cover image, allowing the latter to remain visually accessible.

The encrypted sequence was subsequently embedded into selected middle-frequency wavelet coefficients according to the procedure described in Section 3.3. This arrangement separated watermark-data encryption from frequency-domain embedding. DES was used as the encryption component in the reported implementation; the present experiments evaluated watermark recovery and image quality rather than the cryptographic security of DES.

3.3 Watermark Embedding Strategy

The embedding procedure is described using a grayscale cover image. Colour images were processed using the RGB-channel representation introduced in Section 3.1.3. The watermark was inserted into selected wavelet coefficients after encryption, and the embedding strength was controlled by the parameter optimized through the GA procedure.

The watermark insertion procedure consisted of the following steps.

(1) The original cover image was decomposed using a one-level two-dimensional wavelet transform, producing four subbands: LL, LH, HL, and HH. The LL subband contained the approximation coefficients, while LH, HL, and HH represented detail information in different spatial directions.

(2) The watermark image was converted into a one-dimensional sequence by arranging its pixel values in row-major order. The resulting sequence served as the input for watermark encryption.

(3) The one-dimensional watermark sequence was encrypted using DES to produce the ciphertext sequence used in the embedding operation.

(4) The encrypted watermark sequence was inserted into the wavelet coefficients of the selected LH or HL subband according to Eq. (15).

$$b_{ij} = a_{ij} + \alpha X_{ij} \tag{15}$$

In Eq. (15), a and b denote the wavelet coefficient matrices before and after watermark embedding, respectively. The parameter α controls the embedding strength, while the encrypted watermark sequence provides the information inserted into the selected coefficients.

(5) The embedding-strength parameter α was determined using the GA-based procedure described in Section 3.4. The selected value was used to regulate the modification of the wavelet coefficients while restricting image distortion according to the specified evaluation criterion.

(6) The inverse wavelet transform was applied to the modified coefficients and the remaining wavelet subbands to reconstruct the watermarked image. The resulting image was subsequently used for the imperceptibility and robustness evaluations described in Section 4.

3.4 Determining the Optimal Watermark Embedding Energy

The strength of watermark embedding was controlled by the parameter α introduced in Section 3.3. Increasing this parameter generally increases the modification applied to the selected wavelet coefficients, although the resulting changes in image quality and watermark recovery depend on the host image and the embedding procedure. A suitable value must therefore be determined within the constraints imposed by the selected image-distortion criterion.

In the proposed method, a GA was used to search for an embedding-strength parameter instead of relying exclusively on repeated manual adjustment. Candidate parameter values were encoded and evaluated through an iterative evolutionary procedure. The GA applied selection, crossover, and mutation operations to explore the permitted parameter range. The parameter obtained from this process was subsequently used in the watermark embedding procedure, and its performance was examined through the experiments described in Section 4.

The GA implementation consisted of the following steps.

(1) Initialization. An initial population of candidate embedding-strength values was generated within the prescribed search range. Each candidate represented a possible value of α for watermark insertion.

(2) Encoding. Candidate values of α were represented using binary chromosomes for the GA operations.

(3) Fitness evaluation. The fitness of each candidate was evaluated using the relative-error criterion. In the reported formulation, the fitness function was defined as $F = 1/\mathrm{RE}$. SNR and PSNR were used as supplementary image-quality measures and were not included directly in the fitness calculation.

(4) Genetic operations. Candidate solutions were updated through selection, crossover, and mutation. Random-number generation was used to determine crossover positions and mutation locations, while bit-flipping operations modified chromosome values according to the specified mutation probability.

(5) Parameter selection. The iterative procedure continued until a specified termination condition was reached. The resulting embedding-strength parameter was then used to construct the watermarked image through the procedure described in Section 3.3.

The GA used a population size of 30, a maximum of 400 generations, a crossover probability of 0.7, and a mutation probability of 0.01 [18, 19]. These values were adopted as empirical configuration parameters.

3.4.1 Genetic algorithm termination criteria

The genetic algorithm was configured with a maximum of 400 generations. Candidate embedding-strength parameters were evaluated using the fitness function defined in Section 3.4, and the population was updated through selection, crossover, and mutation. The crossover probability (P_c) and mutation probability (P_m) were fixed at 0.7 and 0.01, respectively, throughout the optimization process.

In the reported implementation, the interval $3\% \leq \mathrm{RE} \leq 5\%$ was adopted as an empirical criterion for selecting the embedding strength. The lower bound was intended to avoid excessively weak embedding, whereas the upper bound restricted the distortion introduced into the cover image. A candidate satisfying this interval was considered acceptable under the specified distortion criterion. If the criterion was not satisfied, the GA continued its search, subject to the maximum generation limit.

The specified RE interval represents a parameter-selection constraint rather than a direct measure of watermark robustness. Although increasing embedding strength may improve watermark recovery under certain conditions, satisfying the RE criterion alone does not establish resistance to image-processing or geometric attacks. The robustness of the selected embedding parameter was therefore assessed separately through the experiments presented in Section 4.

If the maximum generation limit was reached without identifying a candidate that satisfied the prescribed RE interval, the optimization procedure was regarded as unsuccessful under the specified criterion. This outcome should be distinguished from failure of watermark extraction, which was evaluated independently in the subsequent experiments.

3.4.2 Genetic algorithm convergence plot

Figure 1 illustrates the change in relative error during the GA optimization process. The plotted values fluctuated during the initial generations and subsequently decreased as the search progressed. This behaviour indicates that the candidate embedding parameters produced progressively lower reconstruction error under the reported optimization settings. The observed convergence should be interpreted in relation to the specified fitness criterion and does not independently establish improved watermark robustness under attack conditions.

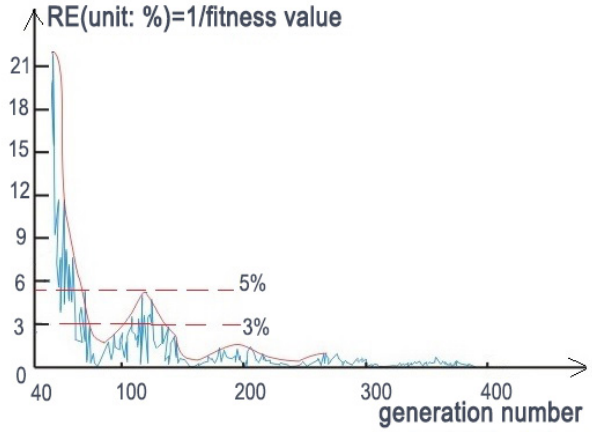


Figure 1. Evolution of relative error during the GA-based embedding-strength optimization

Note: RE = Relative error; GA = Genetic algorithm. Fitness value ($1/\mathrm{RE}$) versus generation number. The blue curve is the original observation curve of RE. The red curve is the smoothing result of the blue curve, which represents the trend.

3.5 Watermark Extraction Procedure

The extraction procedure is described using a grayscale image. Colour images were processed using the RGB-channel representation introduced in Section 3.1.3. Watermark recovery was carried out using the original cover image, the watermarked image, and the embedding-strength parameter determined during watermark insertion.

The extraction process consisted of the following steps.

(1) The original cover image and the corresponding watermarked image were decomposed using the same one-level two-dimensional wavelet transform employed during embedding. The decomposition produced the LL, LH, HL, and HH subbands for both images.

(2) The corresponding wavelet coefficients from the original and watermarked images were used to recover the encrypted watermark sequence according to Eq. (16).

$$X_{ij} = (b_{ij} - a_{ij})/\alpha \tag{16}$$

Here, a and b denote the wavelet coefficient matrices of the original and watermarked images, respectively, and α represents the embedding-strength parameter used during insertion. The extraction operation was performed on the subband selected for watermark embedding.

(3) The recovered ciphertext sequence was decrypted using DES in Cipher Block Chaining (CBC) mode to obtain the watermark data. The decrypted binary sequence was divided into groups of eight bits, and each group was converted into its corresponding pixel value. The resulting one-dimensional pixel sequence was then rearranged into a two-dimensional matrix according to the original watermark dimensions.

The reconstructed matrix represented the extracted watermark and was used for the image-quality and robustness evaluations described in Section 4.

3.6 Watermark Extraction After Attacks

Watermark extraction after image manipulation followed the general procedure described in Section 3.5. For the cropping experiment, the region removed from the watermarked image was filled with the corresponding region of the original cover image before extraction. This reconstruction step restored the image dimensions required for wavelet decomposition and watermark recovery.

The resulting image was then processed using the same wavelet-domain extraction procedure. The recovered watermark was compared with the original watermark to evaluate reconstruction quality. Accordingly, the reported cropping experiment represents an original-image-assisted watermark extraction procedure rather than blind extraction from the cropped image alone.

4 Testing Results and Discussion

The proposed watermarking method was evaluated in terms of visual imperceptibility and resistance to image-processing and geometric attacks. The experiments examined whether the GA-selected embedding strength could preserve image quality while allowing the embedded watermark to remain recoverable after image modification. The evaluation included common image-processing operations, geometric distortions, and mosaic-based presentation attacks. StirMark and 2Mosaic were used to examine the robustness of the watermarking procedure under the corresponding attack conditions.

StirMark provides a testing environment for assessing the response of digital watermarking methods to image transformations, including geometric distortions. The 2Mosaic program was used to examine a presentation attack in which a watermarked image is divided into smaller components. Such attacks are relevant to digital content distributed over communication networks, where images may be modified, reformatted, or divided into separate files during transmission and presentation. Although the tested transformations do not cover every possible attack on digital watermarking systems, they provide experimental evidence of watermark recoverability under several representative conditions.

The experiments used the two-dimensional spline wavelet decomposition described in Section 2. The embedding strength selected through the GA procedure was held constant during the reported attack evaluations. This allowed the response of the embedded watermark to different transformations to be examined without readjusting the embedding parameter for each attack. RE was used as the principal measure of image differences, while SRWE, defined in Section 3.1.2, was used to report reconstruction quality. SNR and PSNR were retained as supplementary image-quality criteria. In this study, the term cover image refers to the original image into which watermark information was embedded, while watermarked image denotes the image obtained after the embedding operation.

4.1 Imperceptibility (Invisibility) Test

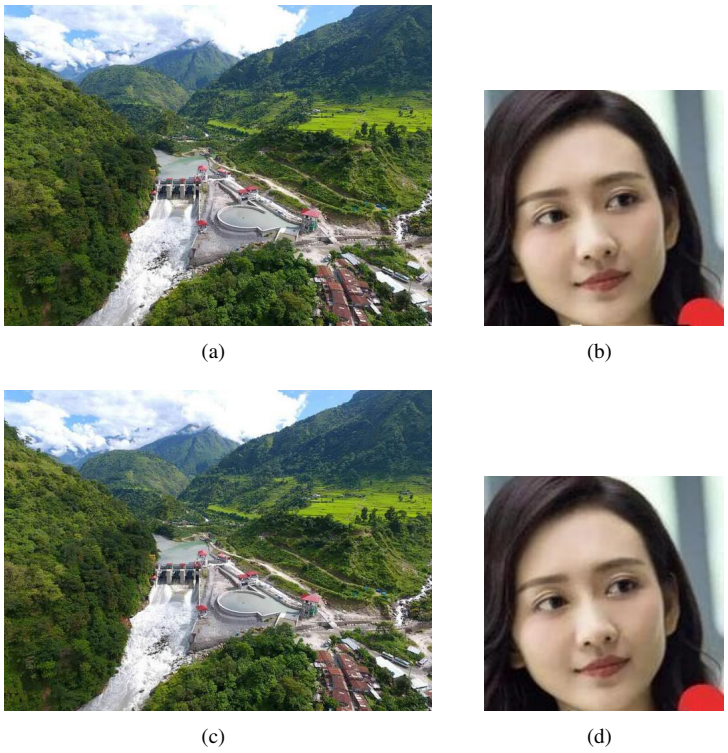
Visual imperceptibility was examined by comparing the original cover image with its watermarked counterpart and assessing the quality of the extracted watermark. Figure 2 presents an example involving two colour images. The original image, designated landscape 1, was used as the cover image, while the face image served as the watermark. Both images were processed through the RGB-channel representation described in Section 3.1.3 before watermark insertion and extraction.

As shown in Figure 2, the watermarked image in Figure 2c retained the main visual characteristics of the original cover image in Figure 2a. No prominent visual distortion was apparent in the displayed example. The watermark extracted from the watermarked image, shown in Figure 2d, also preserved the principal visual features

of the original watermark in Figure 2b. These observations suggest that the selected embedding strength allowed watermark information to be inserted without substantial visible changes to the cover image.

The quantitative evaluation was based primarily on RE, with the other image-quality criteria used as supplementary measures. The RE between the original cover image in Figure 2a and the watermarked image in Figure 2c was 3.6%. The RE between the original watermark in Figure 2b and the extracted watermark in Figure 2d was 2.3%. The first value describes the distortion introduced into the cover image by the watermarking procedure, whereas the second describes the reconstruction error of the extracted watermark.

For the present evaluation, an RE below 5% was treated as an operational criterion for low image distortion. The measured value of 3.6% satisfied this criterion, and the visual comparison in Figure 2 showed no conspicuous changes to the cover image. The extracted watermark also exhibited a relatively small reconstruction error. However, a fixed RE threshold should not be interpreted as a universal measure of human visual perception, since the visibility of image changes depends on image content and the spatial distribution of distortion. Taken together, the numerical results and visual observations indicated that the proposed method maintained acceptable image quality in the reported experiment.



(a) (b)

(c) (d)

Figure 2. Visual assessment of watermark imperceptibility and extraction quality: (a) original cover image (landscape 1); (b) original watermark (face); (c) watermarked image; (d) watermark extracted from (c)

4.2 Robustness Against Common Attacks

Robustness was evaluated by applying image-processing and geometric transformations to watermarked images and examining the information recovered after each attack. A watermark may lose part of its embedded information when the host image is compressed, geometrically transformed, or deliberately manipulated. The ability to recover recognizable watermark information after such operations is relevant to digital ownership verification and the protection of distributed image content [18, 19].

StirMark was used to evaluate the response of the proposed method to representative geometric attacks. The software has been used in digital watermarking research since its introduction in the late 1990s and provides a means of examining how image transformations affect watermark recovery [20]. The following subsections present the reported free-shearing, cropping, and mosaic-attack experiments.

4.2.1 Free shearing attack

A free-shearing attack modifies the spatial arrangement of image information and can disturb the relationship between the locations used for watermark embedding and extraction. This makes shearing a relevant test of watermark

recoverability after geometric distortion. Figure 3 illustrates the response of the proposed method to a free-shearing attack implemented using StirMark 3.1.

In this experiment, the 8-bit grayscale image Lena was used as the cover image, and the 8-bit grayscale image containing the text NCUT served as the watermark. After watermark embedding, the watermarked image was subjected to the reported 10% free-shearing attack. The watermark was then extracted from the transformed image to assess whether the geometric modification had prevented recovery of the embedded information.



(a) (b)

(c) (d)

Figure 3. Watermark extraction after a free-shearing attack using StirMark 3.1: (a) original cover image (Lena); (b) original watermark (sample text "NCUT"); (c) watermarked image after free shearing; (d) watermark extracted from (c)

Note: 10% of the watermarked image (c) is sheared freely.

Figure 3d shows that the extracted watermark retained a recognizable outline of the original NCUT pattern despite the distortion introduced by shearing. Although the recovered image exhibited degradation, the embedded information was not completely lost. This result indicates that the watermark remained partially recoverable under the tested attack condition, rather than failing entirely after the geometric transformation.

Table 1. Success rate of watermark extraction (%) under different attack conditions

Attacks/Method	Proposed GA-method	LSB	Manually Tuned Embedding Energy	Parameters	Notes
1/4 cropping	87.2	9.0	83.1	Remove 25%	See Figure 4
Free shearing	92.6	12.0	87.3	Remove 10%	See Figure 3
Rotation 5°	98.3	28.7	93.4	Angle 5°	
Rotation 15°	93.4	9.2	89.2	Angle 15°	
Rotation 30°	88.1	0.0	84.5	Angle 30°	
Median filter 5×5	92.4	45.8	88.2	Filter 5×5	
Scale up (enlarging)	91.5	34.2	87.2	Double the size	
Scale down (resample)	92.6	35.1	90.4	Reduce by half	
JPEG (Q = 80)	98.2	62.1	94.1	Q = 80	
JPEG (Q = 70)	97.4	27.4	93.2	Q = 70	
JPEG (Q = 50)	96.3	12.0	91.8	Q = 50	
Gaussian noise (10%)	90.1	0.0	86.7	10% (noise intensity/255)	

Note: GA = genetic algorithm; LSB = least significant bit; Q = JPEG quality factor; JPEG = Joint Photographic Experts Group. The Gaussian noise intensity was set to 10% of the maximum pixel value (255).

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The quantitative results in Table 1 provide further evidence of this response. The proposed GA-based method achieved an SRWE of 92.6% under free shearing, compared with 87.3% for manually selected embedding energy

and 12.0% for the LSB baseline. The GA-based method therefore produced a 5.3-percentage-point higher SRWE than manual parameter selection under the same reported condition. The substantially lower value obtained with LSB is consistent with the sensitivity of direct pixel-level embedding to geometric displacement. These observations support the effectiveness of the proposed wavelet-domain embedding procedure against the particular free-shearing attack examined.

4.2.2 Cropping attack

Cropping removes part of the spatial information contained in an image and may consequently destroy watermark data embedded within the removed region. The impact of cropping depends on the distribution of the watermark information, the size and location of the removed region, and the extraction procedure. The present experiment examined watermark recovery after removing one quarter of a watermarked image.

Figure 4 presents the reported cropping experiment. The original cover image, designated beauty 1, was used to carry a watermark depicting a foot. Following watermark insertion, StirMark was applied to crop 25% of the watermarked image. The remaining image therefore contained approximately three quarters of the original spatial content. Watermark extraction was subsequently performed on the attacked image using the procedure described in Section 3.6.

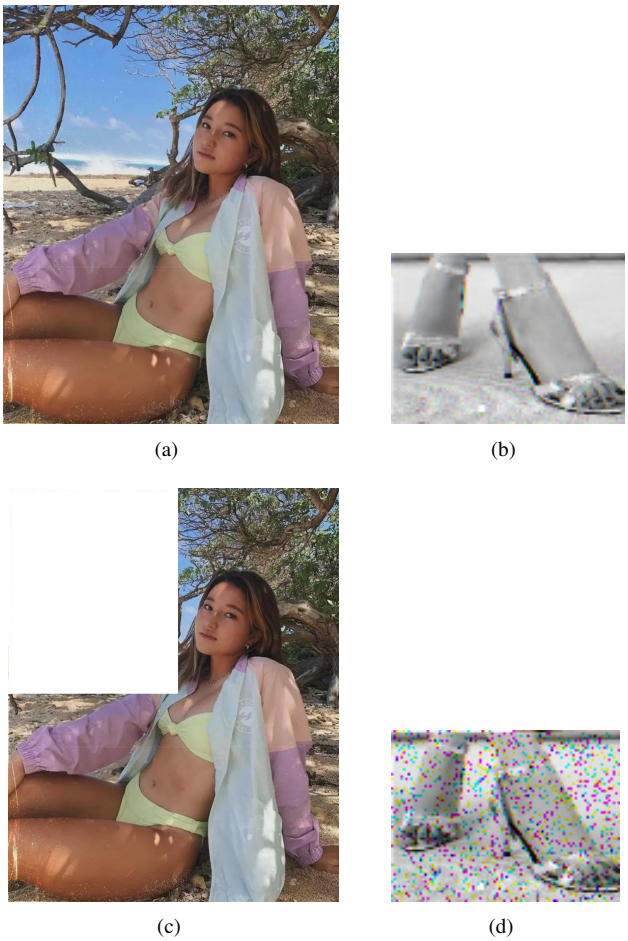


Figure 4. Watermark recovery after a 25% cropping attack: (a) original cover image (beauty 1); (b) original watermark (foot); (c) cropped watermarked image; (d) watermark extracted from (c)
Note: One quarter of the watermarked image (c) is cropped.

The extracted watermark in Figure 4d retained the general outline of the original foot image shown in Figure 4b, although some information was lost during the attack. This indicates that cropping reduced watermark reconstruction quality but did not prevent recovery of its principal visual structure. The result is relevant to watermarking applications in which protected images may be partially cropped before ownership information is examined.

The RE between the original watermark and the extracted watermark was 12.8%, corresponding to an SRWE of 87.2% according to the definition given in Section 3.1.2. The SRWE therefore describes the degree of agreement between the recovered and original watermark under the reported attack, rather than an independently measured

ownership-identification accuracy. Although the reconstruction error was greater than that observed in the attack-free imperceptibility test, the recovered watermark remained visually recognizable.

Table 1 shows that the GA-based method achieved an SRWE of 87.2% after 25% cropping, compared with 83.1% for manually selected embedding energy and 9.0% for LSB. The proposed method therefore exceeded the manually adjusted method by 4.1 percentage points and the LSB baseline by 78.2 percentage points under this condition. The difference between the GA-based and manually adjusted approaches was relatively modest, whereas the difference from LSB was substantially larger.

The observed resistance to cropping is consistent with the distribution of watermark information across selected wavelet coefficients rather than reliance on individual pixel positions alone. Nevertheless, the extent of robustness also depends on how the embedding and extraction operations handle information lost during cropping. The reported experiment demonstrates partial watermark preservation after removing 25% of the image, but a single cropping level does not establish a complete graceful-degradation relationship across increasing attack severity.

The StirMark evaluation included the following example commands:

```
stirmark -Tbase test.ppm
stirmark -S34 -Tbase test.ppm
```

These commands are retained from the reported experimental procedure. Other operational settings were not provided in the original description. A complete reproduction of the experiment would require the corresponding input images, software configuration, and precise attack parameters.

4.2.3 Mosaic attack (presentation attack)

Mosaic attacks represent a form of presentation-level manipulation in which a watermarked image is divided into smaller image components that are subsequently displayed together. Unlike attacks that necessarily alter the pixel values of an entire image, a mosaic attack may exploit the distinction between the visual appearance of content and the individual image files examined by a watermark detector. Under certain presentation conditions, separate image components can be arranged to reproduce the appearance of the original image while interfering with watermark detection performed on individual files [16]. Such attacks are relevant to digital content protection because an image may remain visually recognizable even when its underlying presentation structure has been altered.

The 2Mosaic program was used to examine this type of attack. In the reported implementation, the program divided a watermarked image into smaller subimages that could be displayed together within a web page. This approach is associated with presentation attacks designed to interfere with conventional watermark detection while retaining the overall visual appearance of the protected content [21, 22]. Figure 5 illustrates the corresponding experiment.

In this experiment, the image landscape 2 served as the original cover image, and a text image was used as the watermark. The watermark was embedded using the proposed wavelet-domain procedure before the resulting image was processed using 2Mosaic. The attack divided the watermarked image into 10 smaller subimages, which were subsequently arranged for presentation within a web page. The attack was intended to disrupt conventional detection by changing the image structure encountered by the detector while preserving the overall appearance of the displayed content.

The minimum size of the resulting subimages is an important factor in mosaic-based presentation attacks. When an image is divided into sufficiently small components, individual subimages may contain too little embedded information for reliable watermark detection. A watermarking method intended to resist such attacks must therefore be considered in relation to the spatial distribution of embedded information and the minimum image region from which the watermark can be recovered.

In the reported experiment, the watermark extracted after the mosaic attack remained visually similar to the original text watermark. As shown in Figure 5d, the principal features of the original watermark were retained despite the presentation-level modification. The RE between the original and extracted watermark images was 3.5%, corresponding to an SRWE of 96.5%. This value indicates relatively small reconstruction error under the reported experimental configuration.

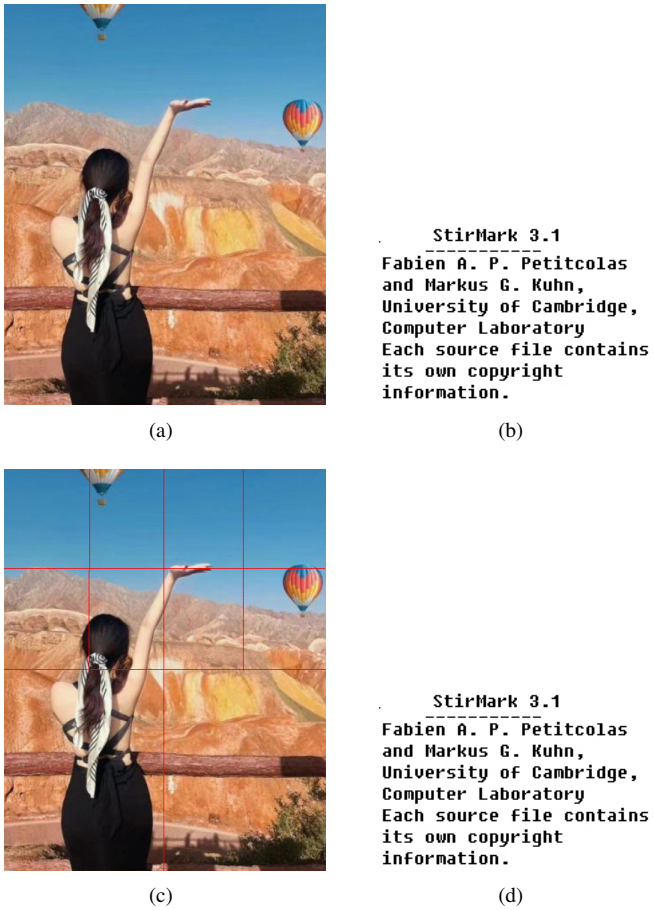
The observed result suggests that the proposed wavelet-domain embedding procedure retained sufficient watermark information for recovery in the tested mosaic configuration. Distributing embedded information across selected wavelet coefficients may contribute to this behaviour, although the available experiment does not isolate the effect of coefficient distribution from other implementation details. The result therefore supports watermark recoverability under the reported 10-component presentation attack, rather than establishing resistance to every possible mosaic configuration.

The following command was reported for the 2Mosaic experiment:

2mosaic 90 testcovert.jpg

Other command settings were omitted from the original experimental description. Reproduction would require additional information about the input image, subimage arrangement, software configuration, and watermark extraction procedure.

The extracted watermark in Figure 5d retained a high degree of similarity to the original watermark in Figure 5b. However, the relationship between RE and visual perception depends on image content and the distribution of reconstruction errors. An RE below 5% was treated as an operational criterion for low distortion in this study, rather than a universally established perceptual threshold [1, 23–25].



(a) (b)

(c) (d)

Figure 5. Watermark extraction under a 2Mosaic presentation attack: (a) original cover image (landscape 2); (b) original text watermark; (c) watermarked image subjected to the 2Mosaic attack; (d) watermark extracted from (c)

4.3 Discussion and Analysis

The experiments provided an initial assessment of the proposed watermarking method under several image-processing and geometric attack conditions. The evaluation included 25% cropping, free shearing, rotation, median filtering, image enlargement and reduction, JPEG compression, and Gaussian noise. These transformations represent different mechanisms of watermark degradation. Geometric attacks alter the spatial arrangement of image information, whereas filtering, compression, and noise affect image coefficients or pixel values. Their combined evaluation allowed the response of the watermarking method to be examined under different forms of image modification.

Table 1 summarizes the reported results for the proposed GA-based method, manually selected embedding energy, and the conventional LSB baseline. All values are expressed as percentages of SRWE, calculated from the relative error between the original and extracted watermarks according to the definition in Section 3.1.2. Higher SRWE values indicate closer agreement between the original and reconstructed watermark images. The comparison therefore concerns watermark reconstruction quality under the specified attacks and should not be interpreted as a direct measurement of copyright-identification accuracy.

As shown in Table 1, the proposed GA-based method achieved higher SRWE values than both comparison methods under all 12 reported attack conditions. The improvement over manually selected embedding energy ranged from 2.2 to 5.3 percentage points, with an average difference of approximately 4.1 percentage points. This suggests

that GA-based parameter selection produced a modest but consistent improvement in watermark reconstruction quality within the tested configurations. The difference from the LSB method was considerably larger, particularly for cropping, rotation, and Gaussian noise.

The improvement over manually selected embedding energy indicates that the automatic selection procedure was able to identify embedding settings associated with higher watermark recovery under the tested attacks. However, these results alone do not establish that the GA reached a global optimum or that equivalent improvements would occur for all image types. The performance of evolutionary optimization can depend on its parameter settings, initialization, fitness function, and termination criteria. Likewise, although automated parameter selection may reduce repeated manual experimentation, the reported results do not provide sufficient runtime measurements to quantify the computational advantage. A systematic efficiency comparison would require measurements of execution time, optimization iterations, and computational resources under identical experimental conditions.

(1) Rotation attacks

Rotation changes the spatial orientation of an image and may disturb the correspondence between embedding and extraction locations. The LSB method is particularly sensitive to this transformation because watermark information is stored directly in individual pixel values. Image rotation generally involves spatial interpolation and resampling, which can modify these values and interfere with watermark reconstruction.

The proposed method achieved SRWE values of 98.3%, 93.4%, and 88.1% at rotation angles of 5°, 15°, and 30°, respectively. Under the same conditions, the LSB method produced values of 28.7%, 9.2%, and 0.0%. The manually selected embedding-energy method achieved 93.4%, 89.2%, and 84.5%.

These results showed that the proposed method retained a larger proportion of the watermark information than the LSB baseline as the rotation angle increased. Its performance also decreased progressively across the three tested rotation angles, rather than falling immediately to zero. This pattern is consistent with the gradual deterioration of watermark reconstruction quality under increasing geometric distortion.

The frequency-domain distribution of watermark information provides a possible explanation for the observed difference between wavelet-domain and pixel-domain embedding. Nevertheless, resistance to rotation may also depend on geometric alignment or correction procedures applied before extraction. The contribution of any such procedure cannot be determined independently from the available results and should be clarified in the implementation description.

(2) Scaling attacks

Scaling modifies image dimensions and typically requires resampling, which can alter the spatial relationships and numerical values of image pixels. In the present evaluation, scaling was used to examine the response of the watermarking method to image enlargement and reduction, two operations commonly encountered during digital content distribution. These operations may also form part of more complex scanning or reproduction processes, although the present experiments did not directly simulate a complete printing-and-scanning workflow.

When the image dimensions were doubled, the proposed GA-based method achieved an SRWE of 91.5%, compared with 34.2% for LSB and 87.2% for manually selected embedding energy. When the image dimensions were reduced by 50%, the corresponding values were 92.6%, 35.1%, and 90.4%, respectively.

The results showed that watermark information embedded through the wavelet-domain procedure remained more recoverable than information embedded using LSB under both scaling conditions. The difference can be partly attributed to the embedding domains. Direct LSB modification depends on individual pixel values, whereas the proposed procedure inserts watermark information into selected wavelet coefficients associated with image structures at different spatial-frequency ranges.

The observed results are consistent with greater tolerance to the resampling operations involved in scaling. However, the experiments were limited to two scaling configurations. Further evaluation over a wider range of scaling factors would be required to determine how reconstruction quality changes with the severity of image resizing.

(3) JPEG compression attacks

JPEG compression reduces image storage requirements by quantizing transformed image information, with greater information loss generally occurring at lower quality factors. Because lossy compression modifies the frequency characteristics of an image, it represents an important test of watermark robustness during digital distribution.

The proposed GA-based method achieved SRWE values of 98.2%, 97.4%, and 96.3% at JPEG quality factors of 80, 70, and 50, respectively. The corresponding values for LSB were 62.1%, 27.4%, and 12.0%, while manually selected embedding energy produced 94.1%, 93.2%, and 91.8%.

The results indicated that the proposed method maintained relatively high watermark reconstruction quality as the JPEG quality factor decreased. At Q = 50, its SRWE remained 96.3%, compared with 91.8% for manual parameter selection and 12.0% for LSB. This suggests that embedding information into selected wavelet subbands was more tolerant of the tested compression settings than directly modifying the LSBs of image pixels.

The difference between the proposed method and manual parameter selection was smaller than the difference between the proposed method and LSB. This distinction is important because it indicates that much of the observed resistance to JPEG compression may be associated with the underlying wavelet-domain embedding strategy, while the GA contributed an additional improvement through parameter selection. Isolating the contribution of GA optimization would require further controlled experiments using identical embedding structures and varying only the parameter-selection procedure.

(4) Gaussian noise attacks

Gaussian noise introduces random perturbations into image pixel values and may interfere with the reconstruction of embedded watermark information. The reported experiment used a noise intensity described as 10% of the maximum 8-bit pixel value of 255. The exact relationship between this intensity and the standard deviation or variance of the Gaussian distribution was not specified in the original experimental description.

Under this attack condition, the proposed GA-based method achieved an SRWE of 90.1%, compared with 86.7% for manually selected embedding energy and 0.0% for LSB. The result indicated that the wavelet-domain methods retained a substantial degree of watermark reconstruction quality, whereas the LSB method failed to recover a comparable watermark according to the reported metric.

The response of the proposed method is consistent with the relative stability of watermark information distributed across selected wavelet coefficients under the tested noise condition. However, a single noise-intensity setting is insufficient to establish the full relationship between noise severity and watermark reconstruction quality. Additional experiments using explicitly defined noise distributions and multiple noise levels would be necessary for a more comprehensive robustness assessment.

4.4 Overall Discussion and Limitations

The experimental findings indicate that the proposed GA-based wavelet-domain watermarking method maintained watermark reconstruction quality under the investigated image-processing and geometric attacks. In the reported comparisons, automatic embedding-energy selection consistently produced higher SRWE values than manual parameter adjustment, while both wavelet-domain approaches generally performed better than the conventional LSB baseline. The imperceptibility experiment also showed a relatively small difference between the original and watermarked images, with an RE of 3.6%.

From the perspective of digital content protection, these results are relevant to applications in which identifying information must remain associated with digital images after routine modifications. The ability to recover embedded watermark information following cropping, rotation, resizing, or compression can support technical procedures for ownership identification and copyright verification. Nevertheless, watermark recoverability alone does not establish resistance to deliberate ownership forgery or provide independent proof of authenticity.

The present findings should be interpreted within the scope of the reported experiments. The evaluation used selected images, a limited range of attack conditions, and two comparison methods. The results therefore establish the performance observed under these configurations rather than the general superiority of the proposed method over contemporary watermarking algorithms. A broader evaluation involving diverse image datasets, more recent methods, repeated optimization trials, and detailed computational measurements would be necessary to assess generalizability. Within these limitations, the results support the feasibility of GA-based embedding-energy selection as a means of improving watermark reconstruction quality while preserving the visual characteristics of protected images.

5 Conclusions

This study investigated a GA-based adaptive wavelet-domain image watermarking method for digital content protection. The proposed method combined DES encryption, frequency-selective watermark embedding, and automated adjustment of embedding energy to address the competing requirements of watermark robustness and visual imperceptibility. The encrypted watermark was embedded primarily into the middle-frequency subbands of the DWT, particularly the LH and HL components, while the embedding strength was determined through GA-based parameter optimization. The method was evaluated under common image-processing operations and geometric distortions, with manually selected embedding energy and a conventional LSB method used for comparison.

The experimental results showed that the proposed method maintained watermark recoverability under several types of image manipulation, including StirMark geometric distortions, cropping, and mosaic-based presentation attacks. The reported watermark extraction rates reached 87.2% after 25% cropping, 93.4% after 15° rotation, and 96.3% under JPEG compression with a quality factor of 50. Under the same conditions, the corresponding rates for manually selected embedding energy were 83.1%, 89.2%, and 91.8%, respectively. The method also maintained a watermark extraction rate of 96.5% in the reported mosaic attack experiment. These results indicated that GA-based embedding-parameter selection provided more reliable watermark recovery than manual parameter adjustment under the tested conditions. The proposed method also outperformed the LSB baseline across the attacks reported in the

comparative experiments. In the imperceptibility test, the RE between the original and watermarked images was 3.6%, indicating limited image distortion under the experimental conditions.

The main contribution of this study is the integration of adaptive embedding-energy selection into a wavelet-domain watermarking procedure intended for digital image protection. By replacing repeated manual adjustment with GA-based parameter search, the proposed approach provides a systematic means of selecting embedding strength while retaining the frequency-selective characteristics of wavelet-domain watermarking. The GA-based procedure automated the selection of embedding strength and reduced dependence on manual parameter adjustment. Its computational efficiency relative to conventional parameter-selection methods remains to be established through controlled runtime experiments. From the perspective of digital content protection, the method provides a technical means of embedding ownership-related information while maintaining image quality and watermark recoverability after common modifications. These properties are relevant to copyright verification, digital ownership identification, and the distribution of protected image content.

The present evaluation was limited to the selected test images, attack conditions, and baseline methods. Further experiments involving larger image datasets, more recent watermarking algorithms, repeated optimization trials, and standardized computational measurements are needed to establish the generality of the reported results. The resistance of the method to deliberate watermark forgery and more advanced security attacks also remains to be investigated. Within these limitations, the findings indicate that adaptive embedding-energy selection is a feasible approach to balancing watermark recovery and visual quality in wavelet-domain image watermarking systems.

Author Contributions

Conceptualization, X.L.J.; methodology, X.L.J., S.F.E., Y.H.A., X.M.W., C.Y.G., and S.C.Y.; software, T.L., S.F.E., and C.Y.G.; validation, T.L.; investigation, T.L.; writing—original draft preparation, X.L.J.; writing—review and editing, S.C.Y. All authors have read and agreed to the published version of the manuscript.

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Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

References

[1] K. Soppari and N. S. Chandra, "Automated digital image watermarking based on multi-objective hybrid meta-heuristic-based clustering approach," *Int. J. Intell. Robot. Appl.*, vol. 7, no. 1, pp. 164–189, 2023. https://doi.org/10.1007/s41315-022-00241-3

[2] I. J. Cox, J. Kilian, T. Leighton, and T. Shamoon, "Secure spread spectrum watermarking for images, audio and video," in *Proceedings of 3rd IEEE International Conference on Image Processing*, Lausanne, Switzerland, 1996, pp. 243–246. https://doi.org/10.1109/ICIP.1996.560429

[3] G. Nicchiotti and E. Ottaviani, "Non-invertible statistical wavelet watermarking," in *9th European Signal Processing Conference (EUSIPCO 1998)*, Island of Rhodes, Greece, 1998, pp. 1–4. https://ieeexplore.ieee.org/document/7089556

[4] K. Zeng, K. Chen, W. Zhang, Y. Wang, and N. Yu, "Robust steganography for high quality images," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 9, pp. 4893–4906, 2023. https://doi.org/10.1109/TCSVT.2023.3250750

[5] Q. Liu, J. Ni, and X. Hu, "Robust image steganography against general scaling attacks," in *Proceedings of the 31st ACM International Conference on Multimedia*, Ottawa, ON, Canada, 2023, pp. 8233–8241. https://doi.org/10.1145/3581783.3612267

[6] M. Kutter, "Watermarking resisting to translation, rotation, and scaling," in *Proceedings of SPIE: Multimedia Systems and Applications*, MA, USA, 1998, pp. 423–431. https://doi.org/10.1117/12.337432

[7] A. M. Abdelhakim, M. H. Saad, M. Sayed, and H. I. Saleh, "Optimized SVD-based robust watermarking in the fractional Fourier domain," *Multimed. Tools Appl.*, vol. 77, no. 21, pp. 27 895–27 917, 2018. https://doi.org/10.1007/s11042-018-6014-5

[8] R. Hu and S. Xiang, "Efficient 2D mapping for reversible data hiding," *IEEE Signal Process. Lett.*, vol. 30, pp. 329–333, 2023. https://doi.org/10.1109/LSP.2023.3256701

[9] P. Garg and R. R. Kishore, "Comparative analysis: Role of meta-heuristic algorithms in image watermarking optimization," in *Proceedings of Second Doctoral Symposium on Computational Intelligence, Advances in Intelligent Systems and Computing*. Singapore: Springer, 2022, pp. 315–327. https://doi.org/10.1007/978-981-16-3346-1_26

[10] Y. Zhang, X. Luo, J. Wang, Y. Guo, and F. Liu, "Image robust adaptive steganography adapted to lossy channels in open social networks," *Inf. Sci.*, vol. 564, pp. 306–326, 2021. https://doi.org/10.1016/j.ins.2021.02.058

[11] F. Hartung and B. Girod, "Fast public-key watermarking of compressed video," in *Proceedings of 1997 IEEE International Conference on Image Processing*, Santa Barbara, CA, USA, 1997, pp. 528–531. https://doi.org/10.1109/ICIP.1997.647966

[12] K. Zeng, K. Chen, W. Zhang, Y. Wang, and N. Yu, "Improving robust adaptive steganography via minimizing channel errors," *Signal Process.*, vol. 195, p. 108498, 2022. https://doi.org/10.1016/j.sigpro.2022.108498

[13] D. Wang, G. Yang, Z. Guo, and J. Chen, "Enhancing adversarial embedding based image steganography via clustering modification directions," *ACM Trans. Multimed. Comput. Commun. Appl.*, vol. 20, no. 1, pp. 1–20, 2023. https://doi.org/10.1145/3603377

[14] W. Lu, J. Zhang, X. Zhao, W. Zhang, and J. Huang, "Secure robust JPEG steganography based on autoencoder with adaptive BCH encoding," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 7, pp. 2909–2922, 2021. https://doi.org/10.1109/TCSVT.2020.3027843

[15] D. Kundur and D. Hatzinakos, "Digital watermarking using multiresolution wavelet decomposition," in *Proceedings of the 1998 IEEE International Conference on Acoustics, Speech and Signal Processing*, Seattle, WA, USA, 1998, pp. 2969–2972. https://doi.org/10.1109/ICASSP.1998.678149

[16] K. N. Hamdy, A. H. Tewfik, T. Chen, and S. Takagi, "Time-scale modification of audio signals with combined harmonic and wavelet representations," in *Proceedings of 1997 IEEE International Conference on Acoustics, Speech, and Signal Processing*, Munich, Germany, 1997, pp. 439–442. https://doi.org/10.1109/ICASSP.1997.599658

[17] S. Mallat and W. L. Hwang, "Singularity detection and processing with wavelets," *IEEE Trans. Inf. Theory*, vol. 38, no. 2, pp. 617–643, 1992. https://doi.org/10.1109/18.119727

[18] V. Parashar and G. Mehta, "PSO and GA parameters for estimation digital watermarking algorithm"," in *2016 5th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO)*, Noida, India, 2016, pp. 383–383. https://doi.org/10.1109/ICRITO.2016.7784985

[19] E. S. M. El-Kenawy, N. Khodadadi, A. Khoshnaw, S. Mirjalili, A. A. Alhussan, D. S. Khafaga, A. Ibrahim, and A. A. Abdelhamid, "Advanced dipper-throated meta-heuristic optimization algorithm for digital image watermarking," *Appl. Sci.*, vol. 12, no. 20, p. 10642, 2022. https://doi.org/10.3390/app122010642

[20] F. A. P. Petitcolas, R. J. Anderson, and M. G. Kuhn, "Attacks on copyright marking systems," in *Information Hiding, Second International Workshop, IH'98*. Berlin, Germany: Springer, 1998, pp. 219–239.

[21] F. A. P. Petitcolas, "Watermarking schemes evaluation," *IEEE Signal Process. Mag.*, vol. 17, no. 5, pp. 58–64, 2000. https://doi.org/10.1109/79.879339

[22] F. A. P. Petitcolas, "'mosaïc' attack," *The information hiding homepage*. https://www.petitcolas.net/watermarking/2mosaic/

[23] X. Yu, K. Chen, Y. Wang, W. Li, W. Zhang, and N. Yu, "Robust adaptive steganography based on generalized dither modulation and expanded embedding domain," *Signal Process.*, vol. 168, p. 107343, 2020. https://doi.org/10.1016/j.sigpro.2019.107343

[24] Y. Zhang, X. Luo, Y. Guo, C. Qin, and F. Liu, "Multiple robustness enhancements for image adaptive steganography in lossy channels," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 8, pp. 2750–2764, 2020. https://doi.org/10.1109/TCSVT.2019.2923980

[25] B. Liu, Y. Feng, P. Stone, and Q. Liu, "FAMO: Fast adaptive multitask optimization," in *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, New Orleans, LA, USA, 2023, pp. 57 226–57 243.